

Rule as Model: the Quadratured Plummer Overlap as the Production Energy

Implementation notes — companion to `mollifiedDerivation.tex`

Abstract

The production energy is *defined* as a fixed Gauss–Legendre rule applied to the exactly integrable inner integral of the Plummer panel. The resulting function of the vertices is real-analytic, exactly invariant under rigid motions, and self-consistent with its own gradient and Hessian as an identity of calculus rather than a numerical achievement. Its deviation from the ideal mollified overlap $A_{\tilde{\eta}}^{\sigma}$ is an exponentially small, smooth deformation of the model, not an error to be driven to zero. The closed-form machinery of the companion note is retained as an offline oracle. All formulas below are verified against complex-step differentiation at $\sim 10^{-16}$; the numbers are collected in §7.3.

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1 The principle

Quasistatic protocols run on the landscape of the energy that is *actually evaluated*. FIRE follows its gradient; Newton solves against its Hessian; an AQS plastic event is an eigenvalue of that Hessian crossing zero; the vibrational spectrum is its spectrum. None of these operations ever consults the ideal integral A_γ^σ — they consult the implemented function. Two properties of the implemented function are therefore load-bearing: *smoothness* (the Hessian must exist and be continuous everywhere the dynamics can go) and *self-consistency* (the force must be the exact gradient of the evaluated energy, and the Hessian the exact derivative of the evaluated force, to machine precision). A third property, *fidelity* — closeness of the implemented function to A_γ^σ — is qualitatively different: it is a statement about which member of a family of smooth models one is simulating, not about whether the simulation is correct.

The companion note achieves all three at once by evaluating A_γ^σ in closed form. The cost is carried by the third property: exact fidelity is what demands the master integral, the Clausen function, the pole–residue bookkeeping, and the near-parallel bridge, the last of which exists solely to make the closed form agree *with itself* in a degenerate corner of parameter space.

Rule as model spends fidelity to buy simplicity. Fix, once and for all, a quadrature rule for the outer (nodal) integration; keep the inner integration exact and elementary; define the production energy to be the rule applied to the inner closed form; obtain forces and stiffness by differentiating *the rule*. The result is a two-parameter model family: σ , the physical softening length, and n , the panel resolution. The deviation

$$\varepsilon_n(\{\vec{v}\}) \equiv \mathcal{A}_n(\{\vec{v}\}) - A_\gamma^\sigma(\{\vec{v}\}) \quad (1.1)$$

is itself a real-analytic function of the vertices, exponentially small in n , sharing every invariance of the ideal model. It deforms the model; it does not corrupt it. The closed form is not discarded: it becomes the oracle against which the rule is certified offline (§6), which is the job a closed form does best.

2 The model

2.1 Panel geometry

For an ordered pair the quadrature nodes ride ∂B and the exact integration runs along ∂A ; the convention fixing which body is which is set once by index order (§2.4) and never by geometry. Edge α of A runs from $\vec{a}$ to $\vec{a}'$, with

$$\vec{e}_\alpha = \vec{a}' - \vec{a}, \quad L_\alpha = |\vec{e}_\alpha|, \quad \hat{t}_\alpha = \vec{e}_\alpha / L_\alpha, \quad \hat{n}_\alpha = (t_{\alpha,y}, -t_{\alpha,x}),$$

$\hat{n}$ outward for CCW vertex order. Edge β of B carries the fixed Gauss–Legendre abscissae ξ_i and weights ϖ_i on $[-1, 1]$, placed at fixed fractions of the edge,

$$\vec{y}_i = \vec{b} + f_i \vec{e}_\beta, \quad f_i = \frac{1}{2}(1 + \xi_i), \quad (2.1)$$

so the nodes are *affine* in the vertices of B . Each node sees edge α through three scalars,

$$\vec{d} = \vec{y}_i - \vec{a}, \quad \tau = \hat{t}_\alpha \cdot \vec{d}, \quad h = \hat{n}_\alpha \cdot \vec{d}, \quad w = \sqrt{h^2 + \sigma^2}, \quad (2.2)$$

its tangential and perpendicular coordinates in the edge frame ($\vec{d} = \tau \hat{t}_\alpha + h \hat{n}_\alpha$) and the softened perpendicular distance, together with the two endpoint offsets

$$u^+ = L_\alpha - \tau, \quad u^- = -\tau. \quad (2.3)$$

Relation to the companion frame. With the companion’s quadratic $Q(s) = s^2 + 2\beta s + \gamma$ along the α parameter, completing the square gives $Q + \sigma^2 = (s - \tau)^2 + w^2$; the identity $w^2 = h^2 + \sigma^2$ is the geometric content of the companion’s panel half-width. The (τ, h, w) frame is the same object with the square already completed, which is what makes every derivative below one line long.

2.2 The primitive and the inner integral

Define

$$\boxed{\Xi(u, w) = u \ln(u^2 + w^2) - 2u + 2w \arctan \frac{u}{w}} \quad (2.4)$$

Then $\partial_u \Xi = \ln(u^2 + w^2)$ — the two non-log terms exist precisely to cancel the derivative of the first — so the inner integral over edge α at node $\vec{y}_i$ is exact and elementary:

$$\mathcal{I}_\alpha(\vec{y}_i) = \int_0^{L_\alpha} \ln(|\vec{x}(s) - \vec{y}_i|^2 + \sigma^2) ds = \Xi(u^+, w) - \Xi(u^-, w). \quad (2.5)$$

Since $w \geq \sigma > 0$, the logarithm’s argument is bounded below by σ^2 and the arctangent’s argument is finite everywhere: (2.5) is a single branch-free expression valid for every configuration, with no near-field, far-field, or parallel special cases.

2.3 The production energy

$$\boxed{\mathcal{A}_n = -\frac{1}{4\pi} \sum_{\alpha \in \partial A} \sum_{\beta \in \partial B} (\hat{n}_\alpha \cdot \hat{n}_\beta) \frac{L_\beta}{2} \sum_{i=1}^n \varpi_i \left[\Xi(u_{\alpha\beta i}^+, w_{\alpha\beta i}) - \Xi(u_{\alpha\beta i}^-, w_{\alpha\beta i}) \right]} \quad (2.6)$$

This is the model. It is not an approximation scheme wrapped around (1.1)’s ideal; it is the Hamiltonian, with n a resolution parameter reported alongside σ . It enters the packing energy exactly where $A_\cap^\sigma$ did, $U_{\text{ov}} = 2 \sum_{A < B} f(\mathcal{A}_n^{AB} / (A_A^t + A_B^t))$ with $f(u) = u^2$ or any other smooth constitutive choice, plus the soft-body springs and the smooth neighbour switch.

2.4 Convention and symmetrization

$\mathcal{A}_n^{A|B}$ (nodes on B) and $\mathcal{A}_n^{B|A}$ (nodes on A) differ by an amount of order ε_n . Fix the convention by particle index — the higher-indexed body of the pair carries the nodes — and never by geometry: a geometry-dependent choice is a branch, and §4.4 explains why branches are forbidden. The symmetrized $\frac{1}{2}(\mathcal{A}_n^{A|B} + \mathcal{A}_n^{B|A})$ restores exact $A \leftrightarrow B$ exchange symmetry of the model at twice the cost; nothing in the physics requires it, since either one-sided form is already a legitimate smooth energy.

2.5 Regularity

On the open set of vertex configurations where every edge has positive length, $\mathcal{A}_n$ is *real-analytic* in all vertex coordinates jointly. The map from vertices to $(\hat{t}, \hat{n}, L)$ is analytic where $L > 0$; the nodes (2.1) are affine; the scalars (2.2) are polynomial in analytic quantities; w is analytic because $h^2 + \sigma^2 \geq \sigma^2$; and Ξ is analytic on $\{u \in \mathbb{R}, w > 0\}$. A finite sum of compositions of analytic maps is analytic. There is no configuration — touching, crossing, parallel, vertex-on-edge — at which (2.6) changes form, because it has only one form. This is stronger than the C^∞ of the ideal model's *evaluation*, which is smooth as a function but piecewise as an algorithm (tiers, bridge); the rule is smooth as an algorithm.

2.6 Invariances and exact conservation

Every quantity in (2.6) is built from dot products of difference vectors of vertices and of frames that co-rotate with them, so $\mathcal{A}_n$ is exactly invariant under simultaneous translation and rotation of both bodies, at every n . Differentiating along the generators: translation invariance gives $\sum_k \partial \mathcal{A}_n / \partial \vec{v}_k = 0$, i.e. pair forces sum to zero; rotation invariance gives $\sum_k \vec{v}_k \times \partial \mathcal{A}_n / \partial \vec{v}_k = 0$, i.e. pair torques balance. Momentum and angular momentum bookkeeping is exact by symmetry, not by convergence; ε_n cannot leak drift or spin into the packing.

3 Derivatives of the rule

The force is the exact gradient of (2.6) and the stiffness its exact second derivative, obtained by the chain rule through (2.4)–(2.2). The pleasant structure is that the transcendental content of all derivatives is *already computed by the energy*.

3.1 Derivatives of the primitive

$$\Xi_u = \ln(u^2 + w^2), \quad \Xi_w = 2 \arctan \frac{u}{w}, \quad \Xi_{uu} = \frac{2u}{u^2 + w^2}, \quad \Xi_{uw} = \frac{2w}{u^2 + w^2}, \quad \Xi_{ww} = -\frac{2u}{u^2 + w^2}. \quad (3.1)$$

The first derivatives are the logarithm and arctangent the energy already evaluated; the second derivatives are rational. Consequently the analytic Hessian of the rule introduces *no new transcendentals* — the same statement the companion proves for the exact tier over several pages holds here by inspection. As a free consistency check, $\Xi_{uu} + \Xi_{ww} = 0$: Ξ is harmonic in (u, w) .

3.2 Geometry Jacobians

All vertex derivatives of the panel scalars reduce to one identity. With $P = I - \hat{t}_\alpha \hat{t}_\alpha^\top$ the projector off the edge direction, and $\vec{d} = \tau \hat{t}_\alpha + h \hat{n}_\alpha$,

$$P \vec{d} = h \hat{n}_\alpha, \quad (3.2)$$

the perpendicular part of the node position. Since $\partial \hat{t}_\alpha / \partial \vec{a} = -P/L_\alpha$ and $\partial \hat{t}_\alpha / \partial \vec{a}' = +P/L_\alpha$, with $\hat{n}_\alpha = R \hat{t}_\alpha$ (R the -90° rotation, $R^\top = -R$), a short computation using (3.2) and $R^\top \vec{d} = h \hat{t}_\alpha - \tau \hat{n}_\alpha$ gives the complete table. On the exact side ($\vec{a}, \vec{a}'$ the endpoints of α):

$$\begin{aligned} \nabla_{\vec{a}} \tau &= -\hat{t}_\alpha - \frac{h}{L_\alpha} \hat{n}_\alpha, & \nabla_{\vec{a}'} \tau &= \frac{h}{L_\alpha} \hat{n}_\alpha, & \nabla_{\vec{a}} L_\alpha &= -\hat{t}_\alpha, \\ \nabla_{\vec{a}} h &= \left(\frac{\tau}{L_\alpha} - 1 \right) \hat{n}_\alpha, & \nabla_{\vec{a}'} h &= -\frac{\tau}{L_\alpha} \hat{n}_\alpha, & \nabla_{\vec{a}'} L_\alpha &= \hat{t}_\alpha. \end{aligned} \quad (3.3)$$

On the nodal side ($\vec{b}$, $\vec{b}'$ the endpoints of β), the affinity of (2.1) gives, with $\bar{f}_i = 1 - f_i$,

$$\nabla_{\vec{b}}\tau = \bar{f}_i \hat{t}_\alpha, \quad \nabla_{\vec{b}'}\tau = f_i \hat{t}_\alpha, \quad \nabla_{\vec{b}}h = \bar{f}_i \hat{n}_\alpha, \quad \nabla_{\vec{b}'}h = f_i \hat{n}_\alpha, \quad \nabla_{\vec{b}}L_\beta = -\hat{t}_\beta, \quad \nabla_{\vec{b}'}L_\beta = \hat{t}_\beta. \quad (3.4)$$

For the normal-normal prefactor, with $c = \hat{n}_\alpha \cdot \hat{n}_\beta$ and $\kappa = \hat{t}_\beta \cdot \hat{n}_\alpha$ (the sine of the inter-edge angle; note $\hat{t}_\alpha \cdot \hat{n}_\beta = -\kappa$),

$$\nabla_{\vec{a}}c = -\frac{\kappa}{L_\alpha}\hat{n}_\alpha, \quad \nabla_{\vec{a}'}c = +\frac{\kappa}{L_\alpha}\hat{n}_\alpha, \quad \nabla_{\vec{b}}c = +\frac{\kappa}{L_\beta}\hat{n}_\beta, \quad \nabla_{\vec{b}'}c = -\frac{\kappa}{L_\beta}\hat{n}_\beta. \quad (3.5)$$

3.3 The assembled panel gradient

Write the single-node panel energy as $E = -\frac{\varpi_i}{8\pi} c L_\beta \mathcal{I}$ with $\mathcal{I} = \Xi(u^+, w) - \Xi(u^-, w)$, and abbreviate the stored transcendentals

$$\Lambda^\pm = \ln((u^\pm)^2 + w^2), \quad \Theta = \arctan \frac{u^+}{w} - \arctan \frac{u^-}{w}.$$

Then, using $u^+ = L_\alpha - \tau$, $u^- = -\tau$, $\nabla w = (h/w)\nabla h$,

$$\nabla \mathcal{I} = \Lambda^+ \nabla L_\alpha - (\Lambda^+ - \Lambda^-) \nabla \tau + \frac{2h\Theta}{w} \nabla h, \quad (3.6)$$

$$\boxed{\nabla E = -\frac{\varpi_i}{8\pi} \left[\mathcal{I} L_\beta \nabla c + \mathcal{I} c \nabla L_\beta + c L_\beta \nabla \mathcal{I} \right]} \quad (3.7)$$

with the ∇ 's drawn from (3.3)–(3.5) for whichever of the four vertices is being differentiated. Every transcendental in (3.6)–(3.7) — $\Lambda^\pm$, Θ — was already evaluated for the energy: *the force costs zero additional transcendental evaluations*, only a few dozen fused multiply-adds per node. Vertex forces accumulate over the two edges adjacent to each vertex of A and of B , and over all panels, exactly as in any edge-pair loop.

Equations (3.3)–(3.7) were checked as a block against complex-step differentiation of the panel energy over randomized geometries: maximum deviation 1.7×10^{-16} (§7.3).

3.4 Consistency is an identity

Because (3.7) is the derivative of (2.6) by calculus, $\vec{F} = -\nabla \mathcal{A}_n$ holds to machine rounding for every configuration and every n . The companion's exact tier had to *earn* this property: its force and energy are separate closed-form evaluations, they disagreed at 10^{-2} near parallel edges until the bridge repaired the cancellation, and even repaired they agree only to $\sim 3 \times 10^{-9}$ across tiers, which then capped the finite-difference Hessian. Under the rule there is nothing to repair and no tier to cross: the self-consistency floor is machine epsilon, the bridge does not exist, and Newton's quadratic convergence runs to the numerical bottom of the *model's* landscape with no consistency ceiling.

3.5 The Hessian of the rule

Second derivatives are mechanical. The second-order geometry maps are, for edge α with $\delta \vec{e}$ the variation of $\vec{e}_\alpha$ ($\pm$ the endpoint variations),

$$d\hat{t} = \frac{P \delta \vec{e}}{L}, \quad d^2 \hat{t}[\delta_1, \delta_2] = -\frac{P \delta_2 (\hat{t} \cdot \delta_1) + \hat{t} (\delta_2^\top P \delta_1) + P \delta_1 (\hat{t} \cdot \delta_2)}{L^2}, \quad d^2 L[\delta_1, \delta_2] = \frac{\delta_2^\top P \delta_1}{L}, \quad (3.8)$$

with $d\hat{n} = R d\hat{t}$ and $d^2\hat{n} = R d^2\hat{t}$. Combining (3.8) with the rational table (3.1) through the product and chain rules yields the exact pair Hessian in one pass over (α, β, i) , reusing the stored $\Lambda^\pm, \Theta$ and adding rational factors only. Two implementation tiers are worth naming. *Tier one, immediate:* complex-step columns of the analytic gradient, $\vec{v}_k \rightarrow \vec{v}_k + ih \hat{e}$, $H_{.k} = \text{Im}(\nabla \mathcal{A}_n)/h$ with $h = 10^{-30}$ — the exact Hessian to machine precision at $2N_v$ gradient passes, with no subtractive cancellation, a drop-in replacement for the finite-difference Hessian while the analytic assembly is written. *Tier two, endgame:* the assembled one-pass Hessian, cost comparable to a single force evaluation, which is also the stiffness matrix for the vibrational analysis.

3.6 The adhesion moment rides along free

The softened membership of A at a node is, in the same frame,

$$\Psi_A^\sigma(\vec{y}_i) = \sum_\alpha \frac{h}{2\pi w} \left[\arctan \frac{u^+}{w} - \arctan \frac{u^-}{w} \right] = \sum_\alpha \frac{h \Theta}{2\pi w}, \quad (3.9)$$

built from the *same* τ, h, w and the *same* two arctangents as the energy panel. The mollified included length of ∂B in A is then the rule applied to (3.9), $\ell_n^\sigma = \sum_\beta \frac{L_\beta}{2} \sum_i \varpi_i \Psi_A^\sigma(\vec{y}_i)$, at a cost of a few fused multiply-adds per node and zero extra transcendentals. The adhesive model $U = f(\mathcal{A}_n/\text{norm}) - \lambda \ell_n^\sigma$ therefore has exact first and second derivatives from the machinery above, with λ a free physical coupling.

4 Deformation, not error

4.1 What ε_n is

By §2.5, $\varepsilon_n = \mathcal{A}_n - A_\cap^\sigma$ is real-analytic, shares every invariance of the ideal model (§2.6), and introduces no length scale beyond those already in the panel geometry. It shifts energies, forces, and stiffnesses by exponentially small smooth amounts: minima of $\mathcal{A}_n$ are true minima of a legitimate Hamiltonian, saddles are true saddles, and AQS event detection is well posed because the Hessian whose eigenvalue crosses zero is the exact Hessian of the energy driving the dynamics. Comparing runs at different n compares slightly different Hamiltonians, so n is fixed within a study and reported with σ .

4.2 What sets the decay rate

The outer integrand $u \mapsto (\hat{n}_\alpha \hat{n}_\beta) \mathcal{I}_\alpha(\vec{y}(u))$ extends analytically off the real interval; Gauss–Legendre error decays geometrically with the rate set by the Bernstein ellipse through the nearest complex singularity. The singularities of $\mathcal{I}_\alpha$ are the branch points of Ξ at $u^\pm = \pm iw$, i.e. they are anchored at the *endpoints of the exact-side edge*, displaced by $\pm i\sigma$ perpendicular to it. The controlling geometry is therefore the proximity of a vertex of A to the nodal edge β , measured in units of σ — the same degenerate direction the companion meets as the arctangent node $u_\star$ and treats with the bridge. Two faces gliding past each other, the configuration a dense packing occupies most, is the *easy* case: no exact-side vertex is near the nodal edge and convergence is essentially instantaneous. The binding case is a vertex grazing an edge; it sets n .

4.3 Worked numbers

Squares, $\sigma = 0.25$, reference the same rule at $n = 200$ –400; single geometry per row, a protocol rather than a guarantee (§6).

Configuration	$n = 8$	$n = 16$	remark
corner contact (vertex near edge)	1.7×10^{-5}	2.0×10^{-8}	1.2×10^{-4} at $n=4$; 2.7×10^{-7} at $n=12$
near-parallel faces, gap $\sigma/2$	2.4×10^{-12}	3.2×10^{-16}	the bridge’s regime; here trivial
sliding corrugation, p2p over 0.2	6.5×10^{-13}	3.2×10^{-16}	41 offsets, faces in registry

The corrugation row is the LS-DEM concern quantified: because the nodes ride the edges analytically and the inner integral is exact, sliding produces no washboard at any resolvable amplitude — the model’s n -dependence under registry motion is at the 10^{-13} level already at $n = 8$.

4.4 Refinement without branches, and the one trap

If a study’s σ and size distribution demand more accuracy at vertex–edge grazing, there are two smooth refinements: raise n (regular, branch-free, the default answer), or use a composite rule whose interior breakpoint *moves smoothly* with the configuration — e.g. split $[0, L_\beta]$ at $L_\beta S(\pi_\beta/L_\beta)$, where π_β is the tangential projection of the nearest A -vertex onto β and S is a smooth sigmoidal clamp into $(0, 1)$. Both preserve analyticity.

The trap is everything else. A hard if on proximity, adaptive n that switches with geometry, a per-pair choice of which body carries nodes made by looking at the configuration, a clamped (min / max) projection: each is a configuration-dependent discrete choice, and each reintroduces exactly the C^0 seams in the energy landscape that the entire mollification program exists to remove — now hidden in the evaluator rather than visible in the geometry, which is worse. The rule’s guarantee (§2.5) is that the evaluator is a single branch-free expression; every branch added for efficiency must be proven configuration-independent or it silently voids the guarantee. This is the one commandment of rule-as-model.

4.5 Limits and reporting

$n \rightarrow \infty$ at fixed σ recovers $A_\cap^\sigma$ spectrally; $\sigma \rightarrow 0$ afterwards recovers the sharp overlap. If a study pushes σ down, n scales like the number of nodes needed to resolve an analyticity width σ against the edge lengths, i.e. $n \propto L/\sigma$ in the worst panel; in practice one certifies at the working σ (§6) rather than trusting an asymptotic constant. Publications report (σ, n) together as the model specification.

5 What this is not

Surface-node discrete element methods (LS-DEM and descendants) also place nodes on boundaries, and their known pathology — forces sensitive to the surface discretization, discontinuities as nodes enter and leave contact — is the natural objection here. The distinction is structural. There, the nodes *define* the contact: each node carries a penetration test, a discrete predicate whose truth value flips as bodies move, so the force law inherits switching at the node spacing and the physics depends on the discretization in a way that does not converge smoothly. Here, the nodes are abscissae of a convergent quadrature applied to a globally smooth, exactly integrated kernel: no node carries a predicate, nothing switches, the energy is one analytic expression at every n , and changing n perturbs the model by e^{-cn} rather than changing the contact logic. The worked corrugation row of §4.3 is the quantitative form of this distinction.

6 Choosing n ; the oracle protocol

The closed form earns its keep as certification. Offline, at the working σ and the study's shape and size distribution: draw randomized pairs weighted toward the configurations AQS visits (faces in registry, vertices grazing edges at separations $\sim \sigma$); evaluate $\mathcal{A}_n$ against $A_\cap^\sigma$ from the exact tier; tabulate $|\varepsilon_n|$ and, by complex-step, $|\nabla \varepsilon_n|$ versus n ; pick the smallest n whose deformation is comfortably below the scales the study resolves (energy drops per plastic event, the stiffness scale of the softest modes tracked). Then freeze n . Production never calls the oracle; the oracle never touches production. The default from §4.3 is $n = 16$; $n = 8$ is defensible when vertex–edge grazing at sub- σ separations is rare for the shapes at hand.

7 Implementation

7.1 Kernel shape

Per edge pair (α, β) : hoist $\hat{t}_\alpha, \hat{n}_\alpha, L_\alpha, \hat{t}_\beta, \hat{n}_\beta, L_\beta, c, \kappa$; loop $i = 1..n$ over nodes: two dot products, one square root, two logarithms, two arctangents, and $\mathcal{O}(30)$ fused multiply–adds for the energy; $\mathcal{O}(40)$ further multiply–adds and *no* further transcendentals for the force (3.7); accumulate eight force components. The loop body is branch-free and identical for every panel, which is the shape a warp wants: one edge pair per lane, transcendentals on the SFUs, segmented reduction (or atomics) into per-vertex forces. Honest accounting against the exact tier: the rule evaluates more transcendentals per pair ($2n$ logs and $2n$ arctangents versus a handful plus one Cl_2), but they are cheap, perfectly regular, and free of the bridge's data-dependent branching; on an SM the regularity dominates, and the force's zero-marginal-transcendental property has no exact-tier analogue.

7.2 Verification protocol

Four checks, in order of power per line of code. (i) Complex-step versus the analytic gradient at randomized configurations — agreement at machine epsilon or the assembly is wrong; this replaces central differences entirely (no cancellation, no step-size tuning). (ii) Invariance sums: total force and total torque per pair vanish to machine epsilon at any n (§2.6); a violation localizes a Jacobian error immediately. (iii) The oracle table of §6. (iv) Spot identities: $\Xi_{uu} + \Xi_{ww} = 0$; $w^2 - h^2 = \sigma^2$.

7.3 Numbers behind this note

Quantity	checked against	agreement
panel gradient (3.3)–(3.7)	complex step, 60 random panels $\times$ 8 coords	1.7×10^{-16}
rule gradient, $n=8$ model, all 16 coords	central difference of the rule	2.7×10^{-10} (CD floor)
Ξ_u, Ξ_w, Ξ_{uw} (3.1)	finite difference	$\sim 10^{-9}$ (FD floor)
Ξ_{uu}, Ξ_{ww} (3.1)	second difference	$\sim 2 \times 10^{-6}$ (FD floor)
$\mathcal{A}_n$ convergence and corrugation	$n=200$ – 400 reference	table of §4.3

7.4 Code map

Suggested layout alongside `energies.py`: `_xi` ((2.4) and its five derivatives, returning the shared $\Lambda^\pm, \Theta$), `_panelScalars` ((2.2)–(2.3) plus the Jacobian table), `_ruleEnergy` ((2.6)), `_ruleForce` ((3.7) with vertex accumulation), `_rulePsi` ((3.9), sharing `_xi`'s arctangents), `_ruleHessian` (tier

one by complex step over `_ruleForce`, tier two assembled). The exact-tier functions `_masterM`, `_tCoreReal`, `_cl2`, `_ceBridge`, `_wBridge` move unchanged into the oracle/test suite.

8 Summary

- The production energy (2.6) is the fixed Gauss–Legendre rule applied to the exact elementary inner integral (2.5): one branch-free expression, real-analytic in the vertices, exactly invariant under rigid motions at every n .
- Forces (3.7) and Hessians are exact derivatives of the rule: self-consistency is an identity, the near-parallel bridge and the tier gap do not exist, and the force adds zero transcendental evaluations beyond the energy’s.
- The gap to the ideal $A_{\cap}^{\sigma}$ is a smooth, exponentially small *model deformation* ε_n , controlled by vertex–edge proximity in units of σ ; faces in registry — the companion’s hardest case — are the rule’s easiest.
- (σ, n) are the model parameters, certified offline against the closed form, which retires from production into the role it is best at: the oracle.
- The one commandment: no configuration-dependent branch anywhere in the evaluator; every refinement must move smoothly with the vertices or not exist.